SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED
CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**
Question Count: 3

Code of Conduct
I Shri Shanmuga Priya K 192324035 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student:

Assessment Tasks - Case Study

1. **Case Study 1: Smart Retail Inventory Management System**

   A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.
2. **Case Study 2: Cloud-Based Document Collaboration System**

   An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.
3. **Case Study 3: Online Food Ordering Platform**

   A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

Assessment Tasks - Case Study

1) Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

A **Smart Retail Inventory Management System** is a cloud-based application that helps retail businesses monitor inventory, sales, and stock movement in real time. Instead of maintaining separate inventory databases at each store, all information is stored in a centralized cloud platform. This enables every branch to access the latest inventory details instantly.

The system follows a **client-server architecture**, where Point of Sale (POS) systems and management dashboards act as clients. They communicate with centralized cloud services through secure **REST APIs**. Cloud computing provides on-demand infrastructure, storage, analytics, and security, making inventory management more efficient and scalable.

Cloud storage stores product information, sales transactions, customer purchase records, and analytics datasets. Platform services perform data analysis and demand forecasting using historical sales data. Security mechanisms such as VPNs, firewalls, API gateways, encryption, and Identity and Access Management (IAM) protect business information from unauthorized access.

Working of the System

When a customer purchases a product, the POS system sends the transaction details to the cloud through REST APIs. The inventory database is immediately updated, ensuring accurate stock information across all stores. The analytics platform processes transaction logs and predicts future product demand. Store managers monitor inventory levels through cloud dashboards and receive alerts when stock becomes low. Because the application runs on cloud infrastructure, additional computing resources can be allocated automatically during festive sales or peak shopping periods.

Architecture Components

1. Client Layer

POS Systems, Store Dashboard, Mobile Management App

2. API Layer

REST APIs, API Gateway, Authentication Services

3. Cloud Platform Services

Inventory Management Service, Analytics Engine, Demand Forecasting, Notification Service

4. Cloud Storage

Product Database, Sales Records, Transaction Logs, Analytics Dataset

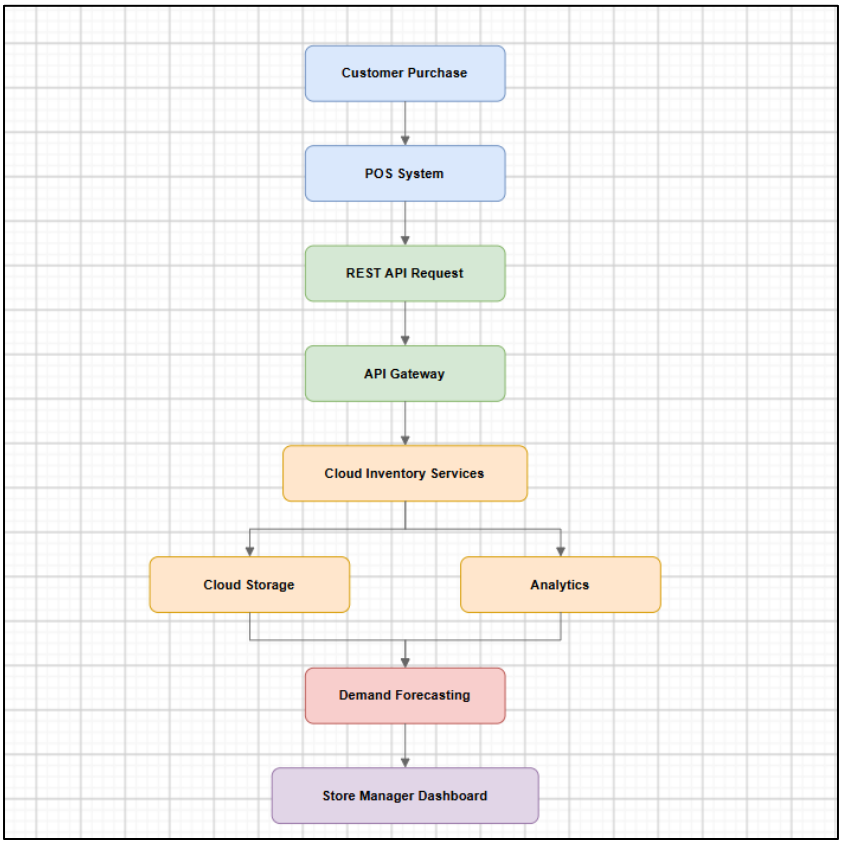
5. Infrastructure Layer

Virtual Machines, Load Balancer, Database Servers, Backup Servers

6. Security Layer

VPN, Firewall, API Gateway, IAM, Encryption

Flowchart



Advantages

- Real-time inventory monitoring, Automatic stock updates
- Centralized inventory management, Improved demand forecasting
- Reduced operational cost, High scalability
- Better customer satisfaction, Secure cloud communication

Conclusion

The Smart Retail Inventory Management System combines cloud infrastructure, storage, analytics, and security services to improve inventory accuracy and operational efficiency. The centralized architecture enables businesses to manage multiple stores effectively while reducing costs and improving decision-making.

2) Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

A **Cloud-Based Document Collaboration System** is a Software as a Service (SaaS) application that enables multiple users to create, edit, and share documents over the Internet. Popular examples include Google Docs and Microsoft Office Online. Documents are stored in distributed cloud storage, allowing users to access their files from any location using web browsers or mobile devices.

The system uses cloud infrastructure, distributed databases, and web APIs to synchronize document updates in real time. Version control keeps track of every change made to a document, allowing users to restore previous versions whenever required. Cloud networking and Content Delivery Networks (CDNs) reduce latency and improve performance for users located around the world.

Security mechanisms such as encryption, identity management, access permissions, and Multi-Factor Authentication (MFA) ensure that only authorized users can view or edit documents.

Working of the System

A user logs into the collaboration platform through a browser or mobile application. After authentication, the requested document is retrieved from distributed cloud storage. Whenever a user edits the document, the changes are transmitted through REST APIs to the collaboration service. The synchronization engine immediately updates all connected users. Version control stores every modification, ensuring data recovery in case of accidental deletion or editing errors.

Architecture Components

1. Client Layer

Web Browser, Mobile App, Desktop Application

2. Web API Layer

REST APIs, Synchronization APIs, Authentication APIs

3. Cloud Platform Services

Collaboration Engine, Synchronization Service, Version Control, Notification Service

4. Cloud Storage

Document Repository, Version History, User Metadata

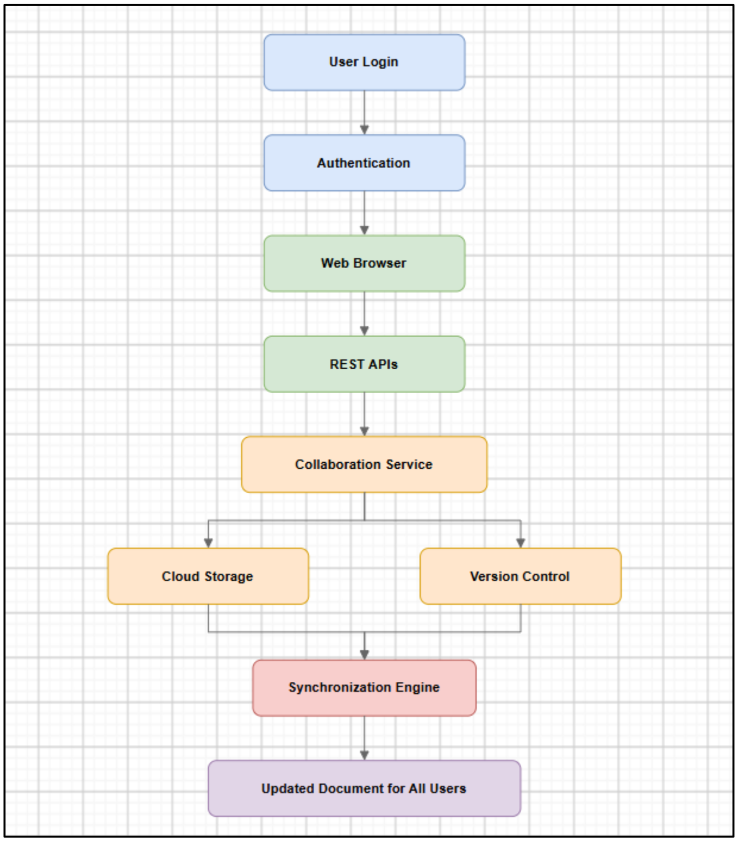
5. Network Infrastructure

CDN, Load Balancer, Distributed Data Centers

6. Security Layer

Identity Management, Access Permissions, Encryption, Multi-Factor Authentication

Flowchart



Advantages

- Real-time collaboration
- Version control
- Automatic synchronization
- Global accessibility
- High availability
- Secure document sharing
- Easy scalability
- Reduced infrastructure cost

Conclusion

A cloud-based document collaboration system enables users to work together efficiently regardless of location. Cloud storage, distributed infrastructure, APIs, and strong security mechanisms provide reliable, scalable, and secure document collaboration.

3) Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

An **Online Food Ordering Platform** is a cloud-based application that allows customers to browse restaurant menus, place food orders, make online payments, and track deliveries through websites or mobile applications. Cloud computing provides scalable infrastructure, reliable storage, and platform services that support thousands of users simultaneously.

The application follows a **three-tier architecture**, consisting of the presentation layer (web/mobile clients), application layer (backend services), and data layer (cloud storage and databases). RESTful Web APIs connect frontend applications with backend services. Cloud storage securely stores menu images, invoices, customer profiles, and order history.

Virtual Machines and Managed Kubernetes provide automatic scaling and fault tolerance. Security mechanisms such as HTTPS, JSON Web Tokens (JWT), Role-Based Access Control (RBAC), and encryption protect customer information and online transactions.

Working of the System

Customers access the platform through mobile apps or web browsers. Their requests are sent through REST APIs to backend cloud services. The application verifies restaurant availability, processes payments, stores order information, and generates invoices. Kubernetes automatically scales application containers when customer traffic increases. Customers and restaurants receive real-time notifications about order status until delivery is completed.

Architecture Components

1. Client Layer

Mobile Application, Web Browser

2. API Layer

REST APIs, API Gateway, Authentication Service

3. Platform Services

Order Processing, Payment Gateway, Notification Service, Restaurant Management

4. Cloud Storage

Menu Images, Customer Information, Order History, Invoices

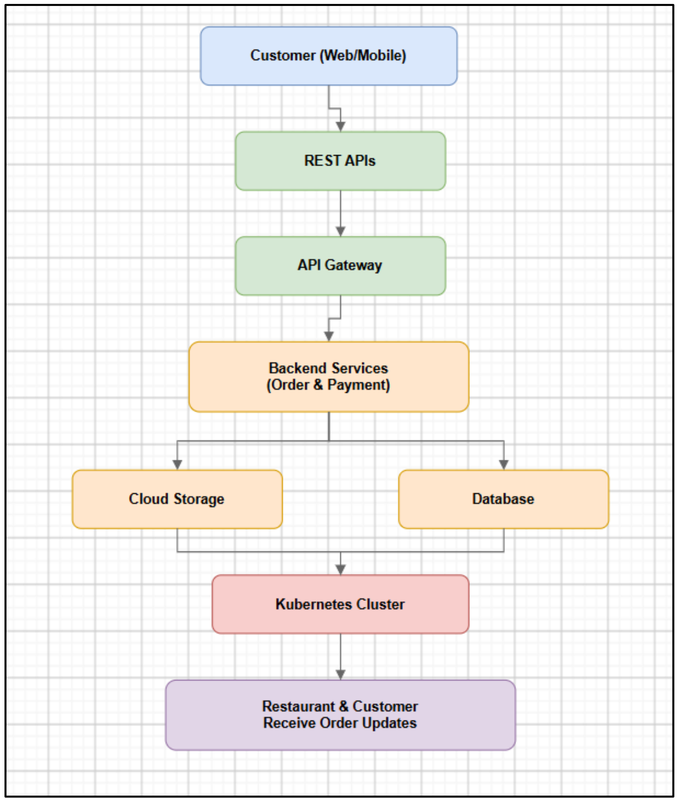
5. Infrastructure Layer

Virtual Machines, Managed Kubernetes, Load Balancer, Cloud Database

6. Security Layer

HTTPS, JWT Authentication, RBAC, Encryption

Flowchart



Advantages

- Convenient online ordering
- Automatic scaling with Kubernetes
- Secure online payments
- High availability, Fast order processing
- Reliable cloud storage
- Better customer experience, Efficient restaurant management

Conclusion

The Online Food Ordering Platform demonstrates how cloud infrastructure, REST APIs, cloud storage, virtual machines, Kubernetes, and security services work together to provide a scalable, secure, and highly available food ordering system. This architecture supports real-time communication, efficient resource management, and excellent customer service.